Diabetes Risk Estimate — Model-Generated Report

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This is a model-generated estimate from an educational project. It is NOT a medical diagnosis and cannot confirm or rule out diabetes. Please consult a qualified healthcare professional for assessment.

Submitted inputs

Field	Value	Unit
Number of pregnancies	1	count
Plasma glucose (2-h OGTT)	150	mg/dL
Diastolic blood pressure	not provided	mm Hg
Triceps skin-fold thickness	not provided	mm
2-hour serum insulin	not provided	mu U/ml
Body mass index	33.5	kg/m²
Diabetes pedigree function	not provided	score
Age	45	years

Model output

The model's estimated probability is 52%, at or above its screening threshold of 29%. The threshold is a model operating point chosen to favour sensitivity, not an established medical cut-off.

Which inputs influenced this estimate

Input	Effect on estimate (percentage points)
Plasma glucose (2-h OGTT)	+20.8
Number of pregnancies	-5.9
Age	+3.8
Diabetes pedigree function	-0.7
Body mass index	+0.5
Triceps skin-fold thickness	-0.1
2-hour serum insulin	-0.1
Diastolic blood pressure	+0.0

Plasma glucose (2-h OGTT) raises the model's estimate by about 21 percentage points; Number of pregnancies lowers the model's estimate by about 6 percentage points; Age raises the model's estimate by about 4 percentage points. These are statistical associations learned by the model from one specific dataset, not causes.

Limitations

The model was trained on 768 records from a single study of adult females of Pima Indian heritage (Arizona, USA, 1990s) and may not generalise to other people. The outcome label follows a glucose-tolerance criterion

whose timing relative to the measurements is undocumented. Probabilities are calibrated to that dataset's prevalence and have not been clinically validated.